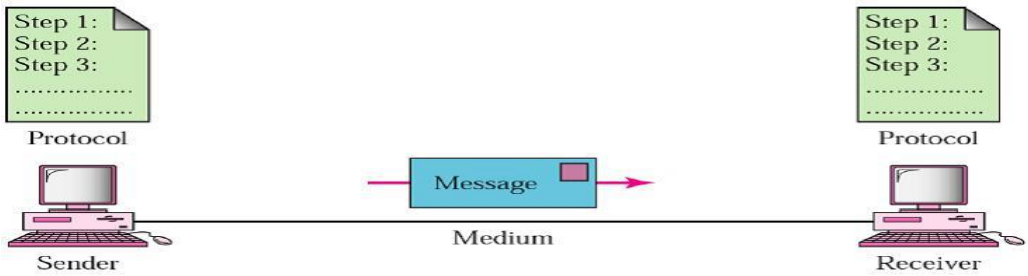
Unit 1: Fundamental of Data Communication

- The term **telecommunication**, which includes telephony, telegraphy, and television, means communication at a distance (tele is Greek for "far").
- The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- The effectiveness of a data communications system depends on **four fundamental characteristics:** delivery, accuracy, timeliness, and jitter.
    - **Delivery**. The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
    - **Accuracy**. The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
    - **Timeliness.** The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.
    - **Jitter.** Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 3D ms. If some of the packets arrive with 3D-ms delay and others with 4D-ms delay, an uneven quality in the video is the result.

Components

- A data communications system has five components.



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- Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
- Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
- Receiver. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
- Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
- Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Data Representation

- Information today comes in different forms such as text, numbers, images, audio, and video
    - **Text:** In data communications, text is represented as a bit pattern, a sequence of bits (0s or 1s). Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding. E.g.: ASCII, Unicode etc.
    - **Numbers:** Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations.
    - **Images:** Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the *resolution.*
    - After an image is divided into pixels, each pixel is assigned a bit pattern. The size and the value of the pattern depend on the image. For an image made of only black and- white dots (e.g., a chessboard), a I-bit pattern is enough to represent a pixel. If an image is not made of pure white and pure black pixels, you can increase the size of the bit pattern to include gray scale.

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- **Audio:** Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete.
- **Video:** Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

Data Flow

- **Simplex:** In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive.
- Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output.
- The simplex mode can use the entire capacity of the channel to send data in one direction.

- **Half-Duplex:** In half-duplex mode, each station can both transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa.
- The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait.
- In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
- Walkie-talkies and CB (citizens band) radios are both half-duplex systems.
- The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

- **Full-Duplex:** In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously.
- The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.

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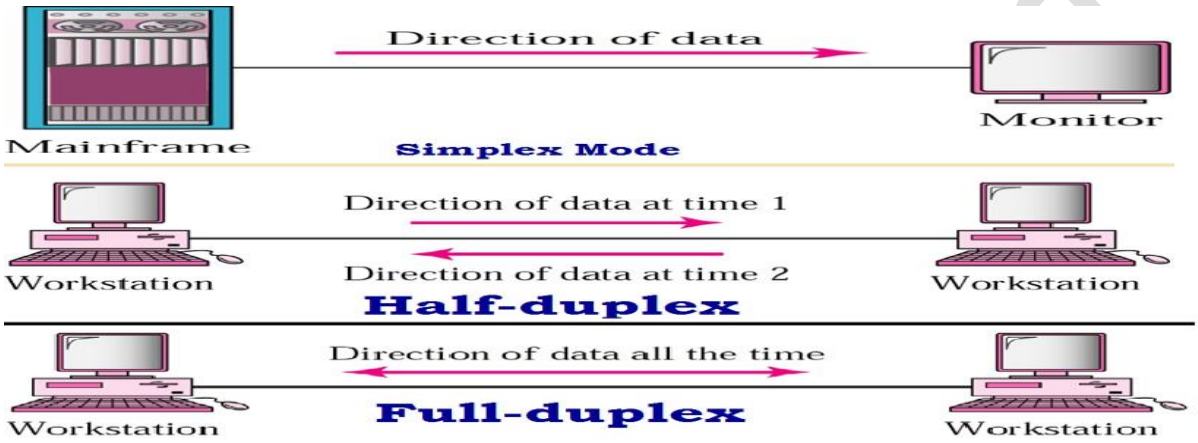
- In full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction.
- This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions. One common example of full-duplex communication is the telephone network.
- When two people are communicating by a telephone line, both can talk and listen at the same time.



Distributed Processing

- Most networks use distributed processing, in which a task is divided among multiple computers.
- Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.

Network Criteria

- A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security.

- **Performance:** Performance can be measured in many ways, including transit time and response time.

- Transit time is the amount of time required for a message to travel from one device to another.
- Response time is the elapsed time between an inquiry and a response.
- The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software.
- Performance is often evaluated by two networking metrics: throughput and delay. We often need more throughput and less delay. However, these two criteria are often contradictory. If we try to send more data to the network, we may increase throughput but we increase the delay because of traffic congestion in the network.

- **Reliability:** In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.
- **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Physical Structures

Type of Connection

- A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.
- There are two possible types of connections: **point-to-point** and **multipoint**.
- **Point-to-Point:** A point-to-point connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.
- When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.
- **Multipoint**: A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link.

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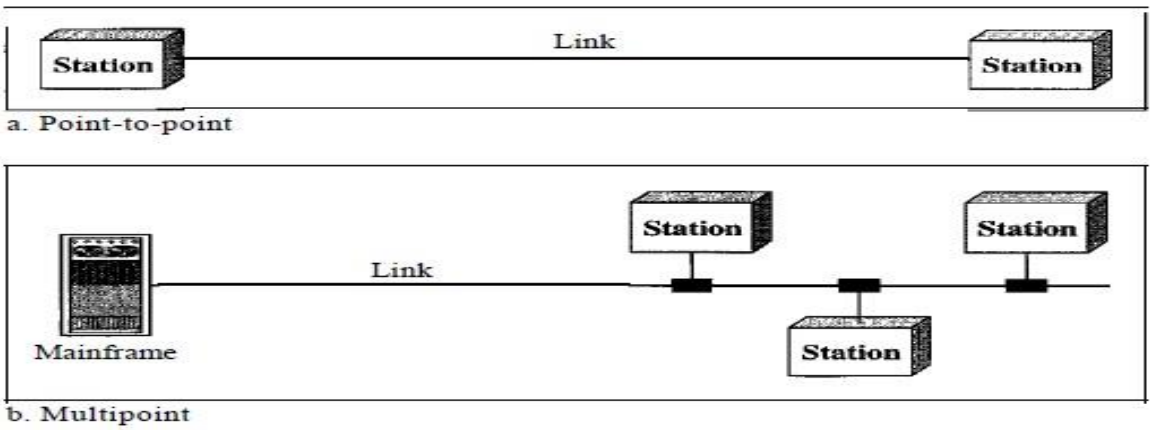
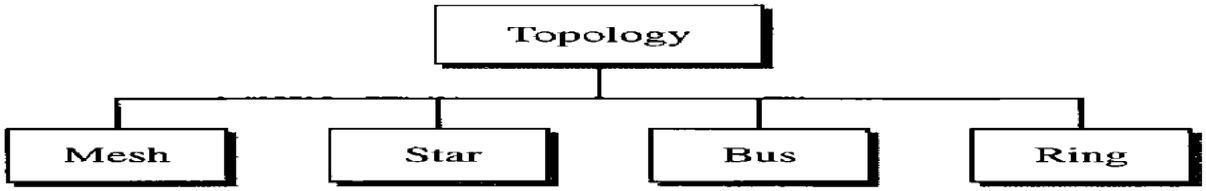
- In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.
- If several devices can use the link simultaneously, it is a *spatially shared* connection. If users must take turns, it is a *timeshared* connection.

Types of connections: point-to-point and multipoint



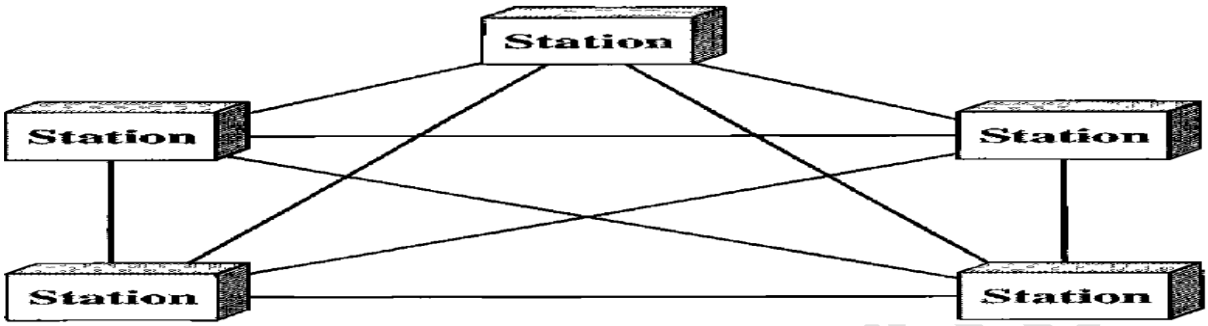
Physical Topology

- The term *physical topology* refers to the way in which a network is laid out physically: two or more devices connect to a link; two or more links form a topology.
- The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus, and ring



- **Mesh Topology:** In a mesh topology, every device has a dedicated point-to-point link to every other device.

- The term *dedicated* means that the link carries traffic only between the two devices it connects. To find the number of physical links in a fully connected mesh network with *n* nodes, we need $n(n-1)/2$ duplex-mode links.



Advantages

- The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
- A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
- There is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it.
- Point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems.

Disadvantage

- The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required.
- First, because every device must be connected to every other device, installation and reconnection are difficult.
- Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
- Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

- **Star Topology:** In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.

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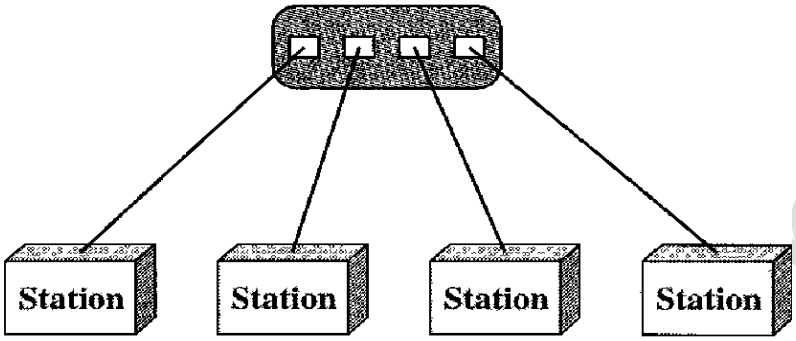
- The devices are not directly linked to one another.
- A star topology does not allow direct traffic between devices. The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.



Advantage

- A star topology is less expensive than a mesh topology.
- In a star, each device needs only one link and one I/O port to connect it to any number of others.
- Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.
- Other advantages include robustness. If one link fails, only that link is affected. All other links remain active.

Disadvantage

- One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

- **The star topology is used in local-area networks (LANs)**

- **Bus Topology:** The preceding topologies all describe point-to-point connections. A **bus topology,** on the other hand, is multipoint.
- One long cable acts as a **backbone** to link all the devices in a network.

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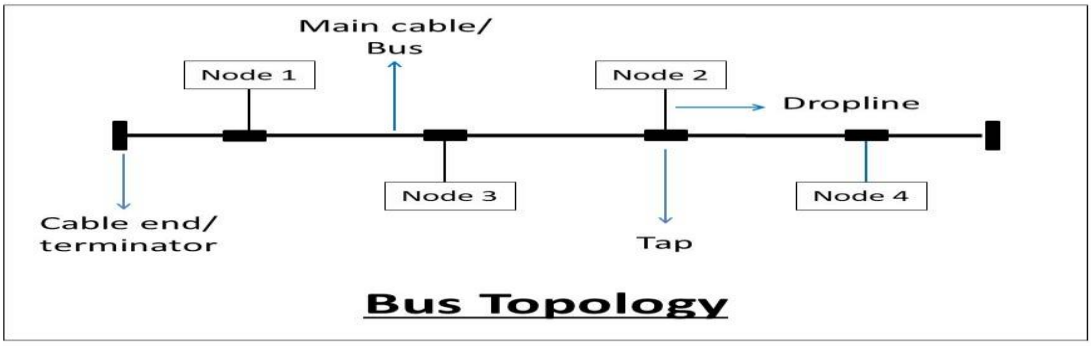

- Nodes are connected to the bus cable by drop lines and taps.
- A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages

- Advantages of a bus topology include ease of installation.
- Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.

Disadvantages

- Disadvantages include difficult reconnection and fault isolation.
- A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.
- A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem.
- The damaged area reflects signals back in the direction of origin, creating noise in both directions.

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- **Ring Topology:** In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along

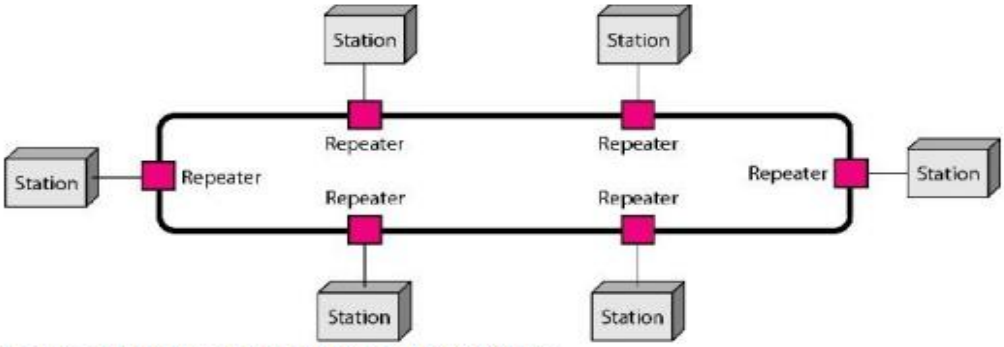

Figure 1.8 A ring topology connecting six stations

Advantage

- A ring is relatively easy to install and reconfigure.
- Each device is linked to only its immediate neighbours (either physically or logically).
- To add or delete a device requires changing only two connections.
- In addition, fault isolation is simplified.
- Generally in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

Disadvantage

- Unidirectional traffic can be a disadvantage.
- In a simple ring, a break in the ring (such as a disabled station) can disable the entire network.
- This weakness can be solved by using a dual ring or a switch capable of closing off the break.

PROTOCOLS AND STANDARDS

Protocols

- A protocol is a set of rules that govern data communications.
- A protocol defines what is communicated, how it is communicated, and when it is communicated.
- The key elements of a protocol are syntax, semantics, and timing.
    - **Syntax:** The term *syntax* refers to the structure or format of the data, meaning the order in which they are presented.
    - For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself.
    - **Semantics:** The word *semantics* refers to the meaning of each section of bits.
    - How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation?
    - For example, does an address identify the route to be taken or the final destination of the message?
    - **Timing:** The term *timing* refers to two characteristics: when data should be sent and how fast they can be sent.
    - For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.

Standards

- Standards are essential in creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing national and international interoperability of data and telecommunications technology and processes.
- Data communication standards fall into two categories: *de facto* (meaning "by fact" or "by convention") and *de jure* (meaning "by law" or "by regulation").
- **De facto**. Standards that have not been approved by an organized body but have been adopted as standards through widespread use are de facto standards.
- De facto standards are often established originally by manufacturers who seek to define the functionality of a new product or technology.

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- **De jure.** Those standards that have been legislated by an officially recognized body are de jure standards.

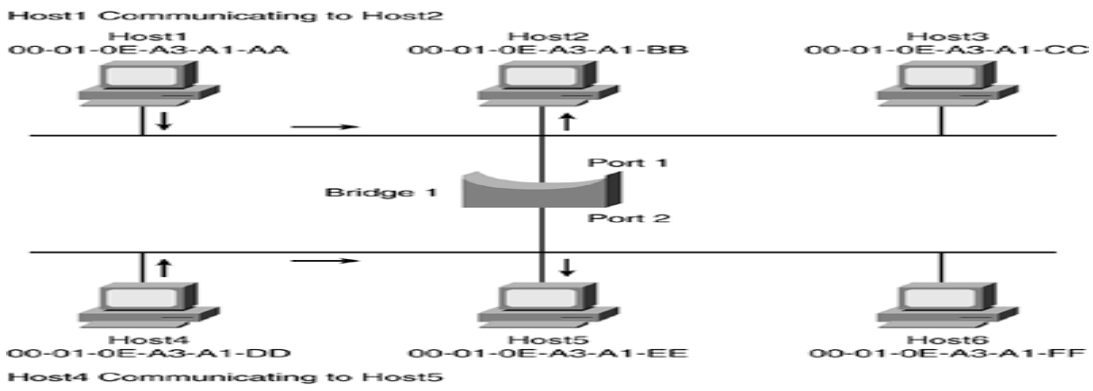
Categories of Networks

H omework

Basic Concepts about: Bridge, Hub, Switch, Router, NIC, MAC-address, Gateway;

Bridge:

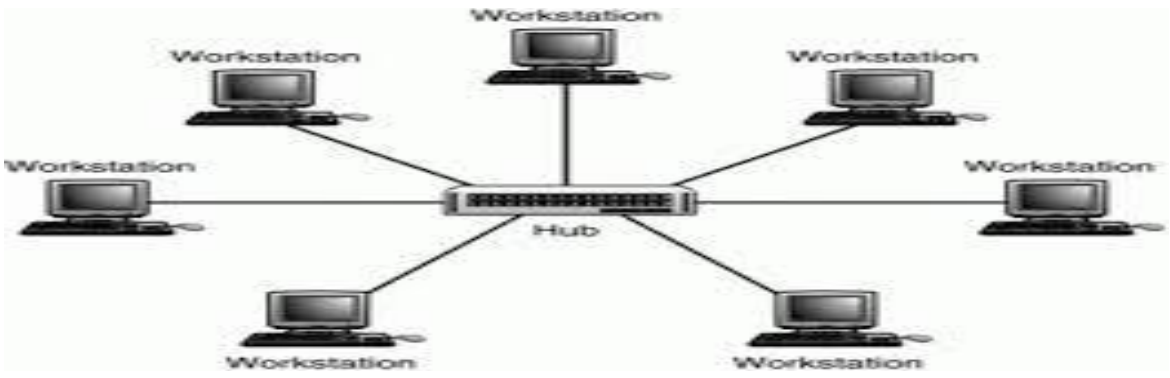
- Bridge is used to divide a large network into smaller segments. Basic functions of the Bridge are the following: –
- Breaking a large network into smaller segments.
- Connecting different media types. Such as connects UTP with the fiber optic.
- Connecting different network architectures. Such as connects Ethernet with the Token ring.
- A Bridge can connect two different types of media or network architecture, but it cannot connect two different types of network layer protocol such as TCP/IP or IPX. Bridge requires the same network layer protocol in all segments.
- **There are three types of Bridge:-**
- **Local Bridge**: – This Bridge connects two LAN segments directly. In Ethernet Implementation, it is known as the Transparent Bridge. In Token Ring network, it is called the Source-Routed Bridge.
- **Remote Bridge**: This Bridge connects with another Bridge over the WAN link.
- **Wireless Bridge**: This Bridge connects with another Bridge without using wires. It uses radio signals for the connectivity.



Hub:

- A hub is basically a multiport repeater.
- Hub is a centralized device that connects multiple devices in a single LAN network. Hubs cannot filter data, so data packets are sent to all connected devices.
- When Hub receives the data signals from a connected device on any of its port, except that port, it forwards those signals to all other connected devices from the remaining ports.
- In other words, collision domain of all hosts connected through Hub remains one. Also, they do not have intelligence to find out best path for data packets which leads to inefficiencies and wastage.
- Usually, Hub has one or more uplink ports that are used to connect it with another Hub.

- **Types of Hub**
- **Active Hub:-** These are the hubs which have their own power supply and can clean, boost and relay the signal along with the network. It serves both as a repeater as well as wiring centre. These are used to extend the maximum distance between nodes.
- **Passive Hub :-** It forwards data signals in the same format in which it receives them. It does not change the data signal in any manner. These are the hubs which collect wiring from nodes and power supply from active hub. These hubs relay signals onto the network without cleaning and boosting them and can’t be used to extend the distance between nodes.



Switch:

- **Switch** is a network device that connects other devices to **Ethernet** networks through **twisted pair** cables.

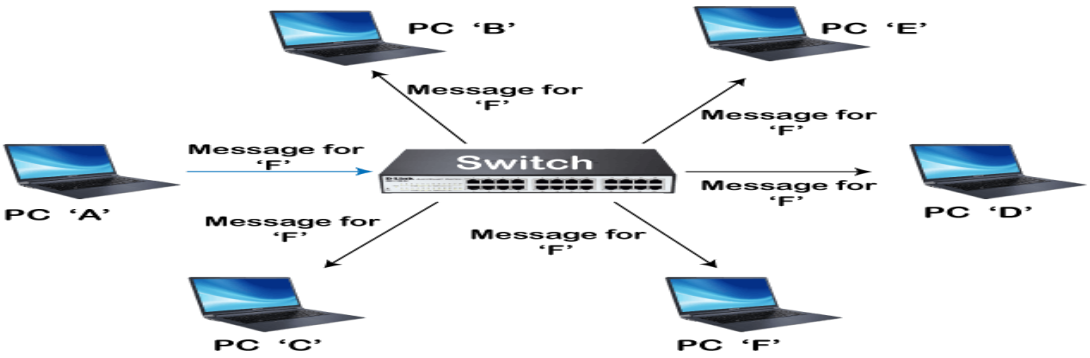
- It uses **packet switching** technique to **receive, store** and **forward data packets** on the network. The switch maintains a list of network addresses of all the devices connected to it.

- On receiving a packet, it checks the destination address and transmits the packet to the correct port. Before forwarding, the packets are checked for collision and other network errors.

- The data is transmitted in full duplex mode.

- Data transmission speed in switches can be double that of other network devices like hubs used for networking.

- This is because switch shares its maximum speed with all the devices connected to it. This helps in maintaining network speed even during high traffic. In fact, higher data speeds are achieved on networks through use of multiple switches.



Router:

- A router is a device like a switch that routes data packets based on their IP addresses.

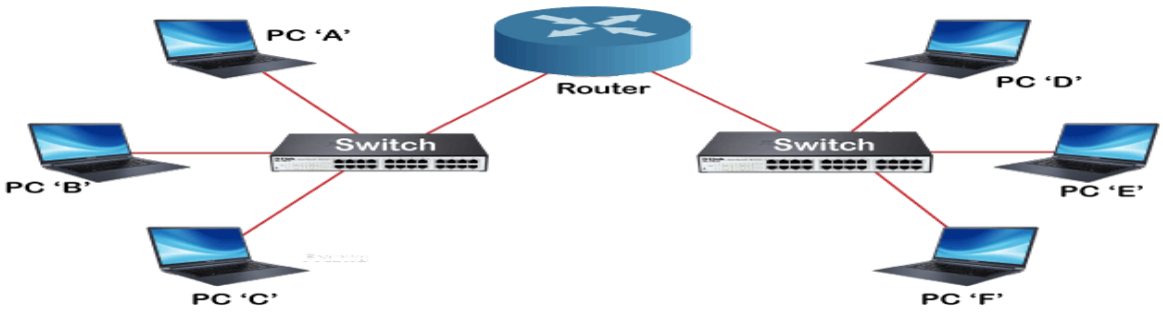
- Router is mainly a Network Layer device.

- Routers normally connect LANs and WANs together if networks support the same set of protocols and have a dynamically updating routing table based on which they make decisions on routing the data packets. Router also strengthens the signals before transmitting them. That is why it is also called **repeater.**

- **Routing Table**
- A router reads its routing table to decide the best available route the packet can take to reach its destination quickly and accurately.

Basically routers are used:

- To connect different network segments.
- To connect different network protocols such as IP and IPX.
- To connect several smaller networks into a large network (known as the Internetwork)
- To break a large network into smaller networks (Known as the Subnets. Usually created to improve the performance or manageability.)
- To connect two different media types such as UTP and Fiber optical.
- To connect two different network architectures such as Token Ring and Ethernet.
- To connect LAN network with Telco company' office (Known as the DTE device).
- To access DSL services (known as the DSL Router).



Connection of networks through Router

NIC:

- NIC stands for Network Interface Card.
- This is also known as network adapter card, Ethernet Card and LAN card. NIC allows a networking device to communicate with the other networking device.
- In modern computer systems, it is an internal hardware component. LAN cards are mounted in different shaped slots on the motherboard.
- Most NIC computers are built-in. LAN cards have a unique code of 48 bits. This unique code is called the MAC (Media Access Control) address.
- This MAC address is stored in the card's ROM. The NIC has RJ45 socket where network cable is physically plugged in.

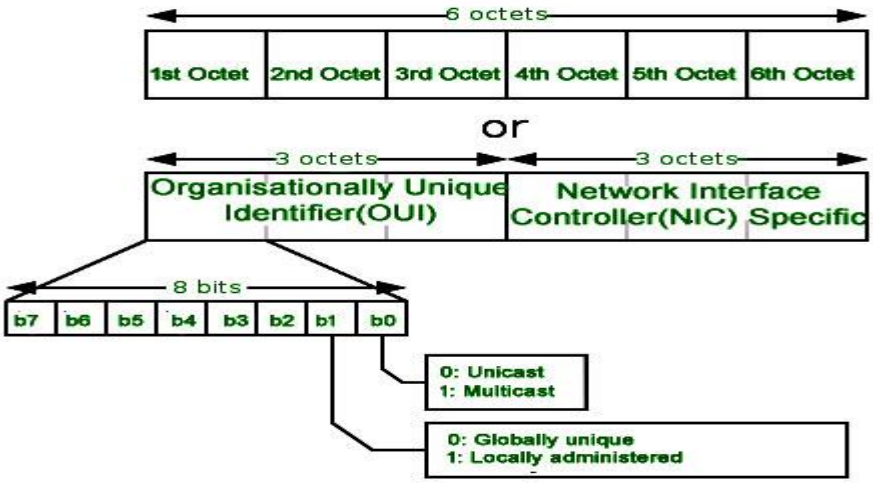
MAC:

- Media Access Control (MAC) address is a physical address that works at the data link layer of the OSI model.

- A MAC address is a 48 or 64-bit address associated with a network adapter.
- MAC addresses are linked to the hardware of the network adapters, hence they are also known as the "hardware address" or "physical address."
- MAC addresses uniquely identify the adapter on the LAN.
- MAC addresses are expressed in hexadecimal notation. For example, "01-23-45-67-89-AB" in a 48-bit address or "01-23-45-67-89-AB-CD-EF" in a 64-bit address. Sometimes, colons (:) are used instead of dashes (-).
- MAC addresses are often considered permanent, but in some conditions, they can be changed.
- There are three types of MAC addresses −
- Unicast MAC Address
- Multicast MAC address
- Broadcast MAC address

MAC Address Format

- A 48-bit MAC address is represented as a string of six octets, "MM:MM:MM:SS:SS:SS".
- The first half (24 bits) of the MAC address contains the ID number of the adapter manufacturer. These IDs are regulated by an Internet standards organization.
- The second half (24 more bits) of the MAC address represents the serial number assigned to the adapter by the manufacturer.



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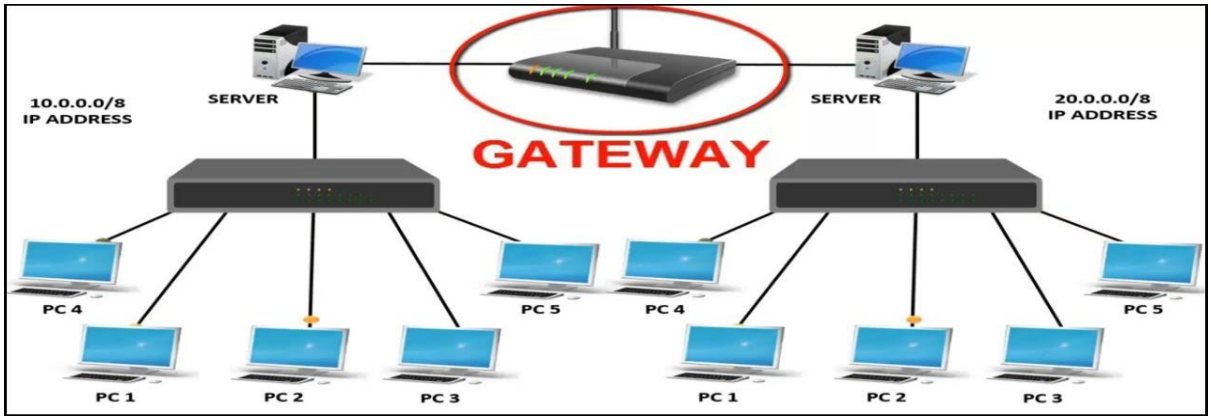
Gateways:

- **Gateway** is a network device used to connect two or more networks having different protocols.

- A gateway usually is a computer with multiple **NICs** connected to different networks.

- A gateway can also be configured completely using software. As networks connect to a different network through gateways, these gateways are usually hosts or end points of the network.

- **Gateway** uses **packet switching** technique to transmit data from one network to another. In this way it is similar to a **router**, the only difference being router can transmit data only over networks that use same protocols.

- Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switch or router.